Xiaojin Zhang

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WORKING EXPERIENCE

08/2023 - Present **Huazhong University of Science and Technology, Wuhan (Assistant Professor)**

08/2021 - 08/2023 **The Hong Kong University of Science and Technology, Hong Kong (Postdoc)**

- **Institute**: Computer Science and Engineering
- **Supervisor:** Prof. Qiang Yang and Prof. Kai Chen

EDUCATION

09/2017 - 08/2021 **The Chinese University of Hong Kong, Hong Kong (PhD)**

- **Institute**: Computer Science and Engineering
- **Supervisor:** Prof. Shengyu Zhang (09/2017 - 06/2019) Prof. Siu On Chan (06/2019 - 08/2021)
- **Major:** Theoretical Computer Science

09/2014 - 07/2017 **The Chinese Academy of Sciences, Beijing (Master)**

- **Institute**: Institute of Computing Technology
- **Supervisor**: Prof. Si-Min He
- **Major:** Computer Application Technology
- **Core Courses:** Advanced Algorithm Design (89/100), The Design and Analysis of Computer Algorithms (89/100), Machine Learning Methods for Computer Vision (95/100)

09/2010 - 07/2014 **Shandong University, Weihai (Bachelor)**

- **Institute:** School of Mathematics and Statistics, Elite Class
- **Dual Degree:** Statistics and Economics
- **Core Courses:** Math Analysis (99/100), Advanced Algebra (98/100), Functional Analysis (92/100), Applied Regression Analysis (99/100), Physics (96/100), Econometrics (96/100)

RESEARCH INTERESTS

Artificial Intelligence, Machine Learning, Federated Learning

PUBLICATIONS (* indicates alphabetic ordering, # indicates equal contributions, ** indicates the corresponding author.)

Theoretical Foundations of Federated Learning and AI

★ **Xiaojin Zhang**, Yahao Pang, Yan Kang, Wei Chen, Lixin Fan, Hai Jin, Qiang Yang. *No Free Lunch Theorem for Privacy-Preserving LLM Inference.* **Artificial Intelligence (AI 2025)** **[CCF A] First Author**

★ **Xiaojin Zhang**, Shuai Li, Shengyu Zhang. *Contextual Combinatorial Conservative Bandits*. **Theoretical Computer Science (TCS 2025)** **[CCF B] First Author**

★ **Xiaojin Zhang**, Yan Kang, Lixin Fan, Kai Chen, Qiang Yang. *A Meta Framework for Tuning Hyperparameters of Protection Mechanisms in Trustworthy Federated Learning.* **ACM Transactions on Intelligent Systems and Technology (TIST 2024)** **[JCR Q1] First Author**

- ★ **Xiaojin Zhang**, Lixin Fan, Siwei Wang, Wenjie Li, Kai Chen, Qiang Yang. *A Game-theoretic Framework for Federated Learning.* **ACM Transactions on Intelligent Systems and Technology (TIST 2024) [JCR Q1] First Author**
- ★ **Xiaojin Zhang**, Yan Kang, Kai Chen, Lixin Fan, Qiang Yang. *Trading Off Privacy, Utility and Efficiency in Federated Learning.* **ACM Transactions on Intelligent Systems and Technology. (TIST 2023) [JCR Q1] First Author**
- ★ **Xiaojin Zhang**. *Improved Algorithm for Testing Permutations.* **Theoretical Computer Science. (TCS 2023) [CCF B] First Author**
- ★ **Xiaojin Zhang**, Hanlin Gu, Lixin Fan, Kai Chen, Qiang Yang. *No Free Lunch Theorem for Security and Utility in Federated Learning.* **ACM Transactions on Intelligent Systems and Technology. (TIST 2022) [JCR Q1] First Author**
- ★ **Xiaojin Zhang**, Honglei Zhuang, Shengyu Zhang, Yuan Zhou. *Adaptive Double-Exploration Tradeoff for Outlier Detection.* **The Thirty-Fourth AAAI Conference on Artificial Intelligence. (AAAI 2020) [CCF A] First Author**
- ★ **Xiaojin Zhang**, Wenjie Li, Shutao Xia, Qiang Yang. *Theoretically Principled Federated Learning for Balancing Privacy and Utility.* **Artificial Intelligence (AI 2025) [CCF A](accept with revision) First Author**
- ★ **Xiaojin Zhang**, Qiang Yang. *Towards Achieving Near-optimal Utility for Privacy-preserving Federated Learning via Data Generation and Parameter Distortion.* **Artificial Intelligence (AI 2025) [CCF A](accept with revision) First Author**
- ★ Pinyan Lu*, Chao Tao*, **Xiaojin Zhang***. *Variance-Dependent Best Arm Identification.* **In Proceedings of the Conference on Uncertainty in Artificial Intelligence. (UAI 2021) [CCF B] Alphabetic Ordering, Equal Contribution**
- ★ Chung-Wei Lee*, Haipeng Luo*, Chen-Yu Wei*, Mengxiao Zhang*, **Xiaojin Zhang***. *Achieving Near Instance-Optimality and Minimax-Optimality in Stochastic and Adversarial Linear Bandits Simultaneously.* **In Proceedings of the 36th International Conference on Machine Learning. (ICML 2021) [CCF A] Alphabetic Ordering, Equal Contribution**
- ★ Jialuo He, Wei Chen, **Xiaojin Zhang****. *Reinforcement Learning as a Catalyst for Robust and Fair Federated Learning: Deciphering the Dynamics of Client Contributions*. **The Thirty-Ninth AAAI Conference on Artificial Intelligence (AAAI 2025) [CCF A] Corresponding Author**

AI Applications and Bioinformatics

- ★ Jiaqi Shao, Tao Lin, Xiaojin Zhang, Qiang Yang, Bing Luo. *Beyond Right to be Forgotten: Managing Heterogeneity Side Effects Through Strategic Incentives.* **The 26th International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc 2025) [CCF B]**
- ★ Yudi Xiong, Yongxin Guo, Weike Pan, Qiang Yang, Zhong Ming, **Xiaojin Zhang**, Han Yu, Tao Lin, Xiaoying Tang. *Camouflaged Variational Graph AutoEncoder against Attribute Inference Attacks for Cross-Domain Recommendation.* **IEEE Transactions on Knowledge and Data Engineering (TKDE 2025) [CCF A]**
- ★ Tao Fan, Hanlin Gu, Xuemei Cao, Chee Seng Chan, Qian Chen, Yiqiang Chen, Yihui Feng, Yang Gu, Jiaxiang Geng, Bing Luo, Shuoling Liu, Win Kent Ong, Chao Ren, Jiaqi Shao, Chuan Sun, Xiaoli Tang, Hong Xi Tae, Yongxin Tong, Shuyue Wei, Fan Wu, Wei Xi, Mingcong Xu, He Yang, Xin Yang, Jiangpeng Yan, Hao Yu, Han Yu, Teng Zhang, Yifei Zhang, **Xiaojin Zhang**, Zhenzhe Zheng, Lixin Fan, Qiang Yang. *Ten challenging problems in federated foundation models*. **IEEE Transactions on Knowledge and Data Engineering (TKDE 2025) [CCF A]**
- ★ Yulin Fei, Yuhui Gao, Xingyuan Xian, **Xiaojin Zhang**, Tao Wu, Wei Chen. *Do Current Video LLMs Have Strong OCR Abilities? A Preliminary Study.* **The 31st International Conference on Computational Linguistics (COLING 2025) [CCF B]**

- ★ Yan Kang, Tao Fan, Hanlin Gu, **Xiaojin Zhang**, Lixin Fan, Qiang Yang. *Grounding foundation models through federated transfer learning: A general framework.* **ACM Transactions on Intelligent Systems and Technology (TIST 2025) [JCR Q1]**
- ★ Dashan Gao, Canhui Wu, **Xiaojin Zhang**, Yuan Yao, Qiang Yang. *Secure Dataset Condensation for Privacy-Preserving and Efficient Vertical Federated Learning.* **European Conference on Machine Learning and Data Mining (ECML PKDD 2024) [CCF B]**
- ★ Qian Chen, Yiqiang Chen, Bingjie Yan, Xinlong Jiang, **Xiaojin Zhang**, Yan Kang, Teng Zhang, Wuliang Huang, Lixin Fan, and Qiang Yang. *Model Trip: Enhancing Privacy and Fairness in Model Fusion across Multi-Federations for Trustworthy Global Healthcare.* **The annual IEEE International Conference on Data Engineering (ICDE 2024) [CCF A]**
- ★ Kangkang Sun, Hansong Xu, **Xiaojin Zhang**, and Kun Hua. *Time-Sensitive Local Differential Privacy-Based Federated Learning for Vehicular Digital Twin Networks.* **International Symposium on Intelligent Computing and Networking. (ISICN 2024)**
- ★ Kangkang Sun, **Xiaojin Zhang**, Xi Lin, Gaolei Li, Jing Wang, Jianhua Li. *Toward the Tradeoffs between Privacy, Fairness and Utility in Federated Learning.* **The 4th International Conference on Emerging Information Security and Applications (EISA 2023)**
- ★ Ming-Qi Liu#, Wen-Feng Zeng#, Pan Fang#, Wei-Qian Cao#, Chao Liu#, Guo-Quan Yan, Yang Zhang, Chao Peng, Jian-Qiang Wu, **Xiao-Jin Zhang**, Hui-Jun Tu, Hao Chi, Rui-Xiang Sun, Yong Cao, Meng-Qiu Dong, Bi-Yun Jiang, Jiang-Ming Huang, Hua-Li Shen, Catherine C. L. Wong, Si-Min He, Peng-Yuan Yang. *pGlyco 2.0 enables precision N-glycoproteomics with comprehensive quality control and one-step mass spectrometry for intact glycopeptide identification.* **Nature Communications**, 8, 438, 2017.
- ★ Wen-Feng Zeng, Yang Zhang, Ming-Qi Liu, Jian-Qiang Wu, **Xiao-Jin Zhang**, Hao Yang, Chao Liu, Hao Chi, Kun Zhang, Rui-Xiang Sun, Peng-Yuan Yang, Si-Min He. *Trends in Mass Spectrometry-Based Large-Scale N-Glycopeptides Analysis[J].* **Progress in Biochemistry and Biophysics**, 2016, 43(6):550-562.

SELECTED HONORS & GRANTS

- **Huawei Donghu Young Scholar**, *RMB 750,000*, 2025.06 - 2028.05, **PI**.
- **Overseas High-caliber Professionals,** *RMB 2,500,000*, 2025.05 - 2031.04, **PI**.
- *"Theoretical and Algorithmic Foundations of High-Utility Privacy-Preserving Federated Learning,"* **Hubei Provincial Natural Science Foundation**, *RMB 100,000*, 2025.06 - 2027.05, **PI**.
- **Domestic-computing-platform-based Ship Motion Prediction Software,** *RMB 600,000*, 2024.12 - 2025.12, **PI.**
- **Research and Practice on Teaching Reform of Bridging Theory and Application in Machine Learning within an Interdisciplinary Context,** *RMB 200,000*, 2024.09 - 2024.12, **Participate.**

SELECTED MANUSCRIPTS

1. **Xiaojin Zhang**. *Near-Optimal Algorithm for Distribution-Free Junta Testing.*
2. Yan Kang, Jiahuan Luo, Yuanqin He, **Xiaojin Zhang**, Lixin Fan, Qiang Yang. *A Framework for Evaluating Privacy-Utility Trade-off in Vertical Federated Learning.*
3. Meijing Liu#, Xinhang Hou#, **Xiaojin Zhang#**, Wei Chen, Zhiheng Chen, Shuang Li, Yi Cui, Daliang Huo, Zekun Cai, Yujie Qin, Yanhua Sun, Qingmin Yang, Chao Liu, Xiaogang Wang. *A Multimodal-reinforced Deep Learning Approach Guided Hunting for Drug Co-assembly and Validation.*
4. **Xiaojin Zhang**, Anbu Huang, Lixin Fan, Kai Chen, Qiang Yang. *Probably Approximately Correct Federated Learning.*
5. Yue Cui, **Xiaojin Zhang**, Xiaofang Zhou, Qiang Yang. *Towards Fair Heterogeneous Federated Learning.*

PATENTS

- **Xiao-Jin Zhang**, Wen-Feng Zeng, Jian-Qiang Wu, Rui-Xiang Sun, Si-Min He. *A comprehensive system for constructing N-glycan databases based on linear canonical representations. (CN108052801B)*
- Wen-Feng Zeng, Ming-Qi Liu, **Xiao-Jin Zhang**, Jian-Qiang Wu, Yang Zhang, Rui-Xiang Sun, Peng-Yuan Yang, Si-Min He. *A comprehensive system for intact glycopeptide identification. (CN106018535A)*

MAJOR AWARDS

- **12/2024** Overseas High-caliber Professionals
- **11/2024** Huawei Donghu Young Scholar
- **2017-2021** Postgraduate Studentship, CUHK
- **12/2019** Travel Award, AAAI
- **06/2017** Excellent Master's Thesis
- **06/2013** The 13th "Challenge Cup" National Undergraduate Curricular Academic Science and Technology Work Competition, Second Prize in College
- **04/2013** The Mathematical Contest in Modeling (MCM), Honorable Mentions
- **12/2012** Youth Entrepreneurship Competition, Third Prize in Shandong Province
- **10/2012** China Undergraduate Mathematical Contest of Modeling (CUMCM), National First Prize
- **09/2012** Summer Social Practice, First Prize at School
- **03/2012** Student Research Training Program, Second Prize at School
- **11/2011** National Scholarship
- **10/2011** The First Prize Scholarship, Outstanding Student in College
- **07/2011** Essay Competition, Second Prize at School
- **05/2011** Aerobic Gymnastics, Second Prize in College

SELECTED TEACHING EXPERIENCE & TALKS

- Organizer and Host, "Trustworthy Machine Learning", Joint Forum of **Huazhong University of Science and Technology - The Hong Kong University of Science and Technology**, 2024
- Course "Machine Learning", **Huazhong University of Science and Technology**, 2024
- Talk "Trustworthy Federated Learning", **Tsinghua University, Xi'an Jiaotong University, Southwest University of Finance and Economics, Nanyang Technological University**, 2023&2024
- Online Talk "Trustworthy Federated Learning: No Free Lunch Theorem", Sep 7, 2022, 4,000 + viewers
- Theory Lunch Talk, **CUHK**, June 22, 2021
- Guest Lecture Assistant, Game Theory (**Yao Class**), **Tsinghua University**, May 10, 2021
- TA, Fundamentals for Embedded Systems (CENG 2030), **CUHK**, Spring 2021
- TA, Techniques for Data Mining (ENGG 5103), **CUHK**, Fall 2020
- TA, Probability and Statistics for Engineers (ENGG 2430), **CUHK**, Spring 2019
- TA, Design and Analysis of Algorithms (CSCI 3160), **CUHK**, Fall 2018
- TA, Approximation Algorithms (CSCI 5160), **CUHK**, Spring 2018
- TA, Design and Analysis of Algorithms (CSCI 3160), **CUHK**, Fall 2017

ACADEMIA & COURSE PROJECTS

Student Research Training Program **03/2011 - 05/2012**

- Responsible for overall planning and allocating duties to team members based on their advantages.

- Applied my professional knowledge to establish logistic and SIR models for the analysis of the micro-blog's potential value. Besides, a misconception was found and rectified through model analysis.
- Composed the final report and was invited to share my modeling experience with the participants in 2013.

External Sort - The First Program I Implemented **08/2014 - 09/2014**

- Given a file containing a large number of double-precision floating-point numbers, e.g., two hundred and fifty million, sort them in ascending order and output them into a new file.
- The loser-tree based merging approach was employed to refrain from the limitation of memory. To further improve the efficiency of the program, the radix sort for double-precision floating-point numbers was implemented.

Construct a Retrieval System **11/2015 - 12/2015**

- Responsible for overall planning and allocating duties to team members based on their advantages. It was impressive that several ideas proposed by me were found to have been introduced in the book, which gave me joy and sorrow at the same time.
- Implemented single-link clustering and k-means clustering separately, analyzed the performance of these algorithms on various corpora, and reduced the complexity of these algorithms effectively.
- Constructed the query vector and document vector, and reduced the dimensions of these vectors by taking advantage of their sparsity. Implemented a text filtering algorithm based on the vector space model. Composed the final report.

Forecast Use of a City Bikeshare System **12/2014 - 01/2015**

- Combined historical usage patterns with weather data to forecast bike rental demand in the Capital Bikeshare program in Washington, D.C.
- Compared the effects of using different data preprocessing methods such as PCA, log transform and zscore, different feature sets, different models including SVM combined with time series, GBRT, and the trained model with the smallest root mean squared logarithmic error on test data was selected.

Build Inverted Indexes **03/2015 - 05/2015**

- Simulated the fragmentation of the proteins into peptides according to the specified enzyme type, and built three index files to facilitate the search for peptide sequences satisfying the filtering criteria.
- Filtered out the redundant peptides based on the loser tree merge and outputted the peptides in ascending order according to their mass value.
- The running time on a given dataset was reduced from almost 900 seconds to 1 second through optimization. As a result, this program was tested to have the highest efficiency.

Create a Protein Search Engine **07/2015 - 08/2015**

- The MS/MS spectra were regarded as sentences, and the proteins were regarded as paragraphs, we need to figure out the matching relations between the sentences and the paragraphs.
- The initially matched peptides were imported based on the index files, and the modified peptides were generated based on the dynamic programming. The peptides were further rated using the kernel-based spectral dot product, and the one with the highest score was selected as the most confident identification.

- Wrote a script using python to compare the spectra identified by my engine and those identified by pFind 2.8, and composed the final report.

Identification of Glycopeptide with Machine Learning Model **11/2015 - 12/2015**

- Read hundreds of spectra and record the corresponding features. Through the combination of the conclusion of data analysis, literature research, and the experience accumulated in our group, more than 100 kinds of features were extracted.
- Tried distinct regression models including decision tree, adaboost, ridge, lasso, svr, and different classification models including random forest, gradient boosting, logistic regression, svm. The testing results showed that the assembled models like adaboost, gradient boosting, and random forest performed better than the other models.

Glycan Database Construction Based on the Topological Structures **09/2015 - 12/2015**

- Took over the codes written by my senior brother, aimed at building a rational as well as comprehensive glycan database, thereby making it more beneficial for the identification of glycopeptides.
- Improve the running time of the code from 650 seconds to 2 seconds.
- Detected the mirrored glycan structures by comparing the structures through a depth-first traversal, and two novel hashing methods were successively proposed to prune the comparison operations efficiently.
- The canonical representations for the glycans were generated. The glycans would be concluded as isomorphic if and only if they correspond to identical canons. As a result, the glycan isomorphism problem could be equivalently recast as the problem of identifying the duplicate numbers.
- Read lots of papers about data mining and implement different canonical representation schemes of glycans. In addition, distinct implementation methods were utilized for the same canonical representation scheme in order to select the most efficient one. The application of canonical representation not only ensured high efficiency in terms of application but also guaranteed the elegance of the algorithm in view of theory.

Glycan Database Construction Based on the Canonical Representations **02/2016 - 04/2016**

- Designed and implemented the algorithms, including enumerating the glycans based on the canonical representations, generating the subglycans of each glycan, and calculating the degree of each glyco according to the canonical representation of the glycan.
- Consequently, the memory requirements have been reduced and the efficiency of the program has been improved to a great extent. Additionally, the conclusion of literature research showed that no better algorithm had ever been put forward.

OTHER EXPERIENCE

Haihua Institute Beijing, China
Research Intern 03/2021-06/2021

Shanghai Qizhi Institute Shanghai, China
Research Intern 09/2020-01/2021

Shanghai University of Finance and Economics Shanghai, China
Summer Course 07/2018-08/2018

Tencent Quantum Lab Shenzhen, China
Research Intern 06/2018-07/2018

Shanghai Jiao Tong University Shanghai, China
Summer Course 07/2017-08/2017

EXTRACURRICULUM EXPERIENCE

- Interested in playing the piano, dancing, and doing yoga.
- Served as a volunteer in HOBIE and attended the etiquette training which was really beneficial.
- Took the duty of broadcasting Image-Text, invited by the communities as a media reporter.
- Succeeded in getting support from some relevant departments during the Summer Social Practice by virtue of perseverance.
- Organized and participated in various dance shows with my classmates.
- Cooperated with four classmates to translate Philip Guo's memoir--The Ph.D. Grind--into Chinese.

SKILLS

- CET4 (556), CET6 (511), GRE (152+165+3.5), TOEFL (92, writing 29)
- Have experience with C++, Matlab, Lingo, Eviews, SPSS, R and Python.